Pandas High-Frequency Interview Questions

Data Loading

read_csv(), read_parquet(), dtype, parse_dates

Datetime

to_datetime(), strftime(), .dt.year/.month/.day/.quarter, invalid dates

Filtering

loc vs iloc, boolean masks, isin

GroupBy

groupby(), agg(), transform(), size() vs count()

Maximum Rows

idxmax()+loc, transform('max'), ties

Merge

merge vs concat, inner/left/right/outer, validate

Missing Values

isna(), fillna(), dropna(), ffill(), bfill()

Duplicates

duplicated(), drop_duplicates(), keep latest

Sorting

sort_values(), nlargest(), rank()

Time Series

shift(), diff(), pct_change(), rolling()

Performance

vectorisation, avoid apply(axis=1), production code

Production

schema validation, dtypes, edge cases

Most Common Live Coding Tasks

- Highest volume per year
- Highest close per year (keep all ties)
- Merge reference tables
- Daily/monthly aggregation
- Parse dates
- Remove duplicates
- Handle missing values
- Rolling averages
- Previous value using shift()
- Returns using pct_change()

Typical Follow-up Questions

- Why groupby?
- Why not sort_values()?
- Why idxmax()?
- Why transform()?
- What if there are duplicate maximum values?
- How would you optimise this?
- How would you test hidden cases?

Pair Coding Checklist

1. Clarify requirements.
2. Inspect schema (head, columns, dtypes).
3. Convert data types.
4. Explain approach.
5. Implement.
6. Discuss edge cases.
7. Mention complexity and trade-offs.